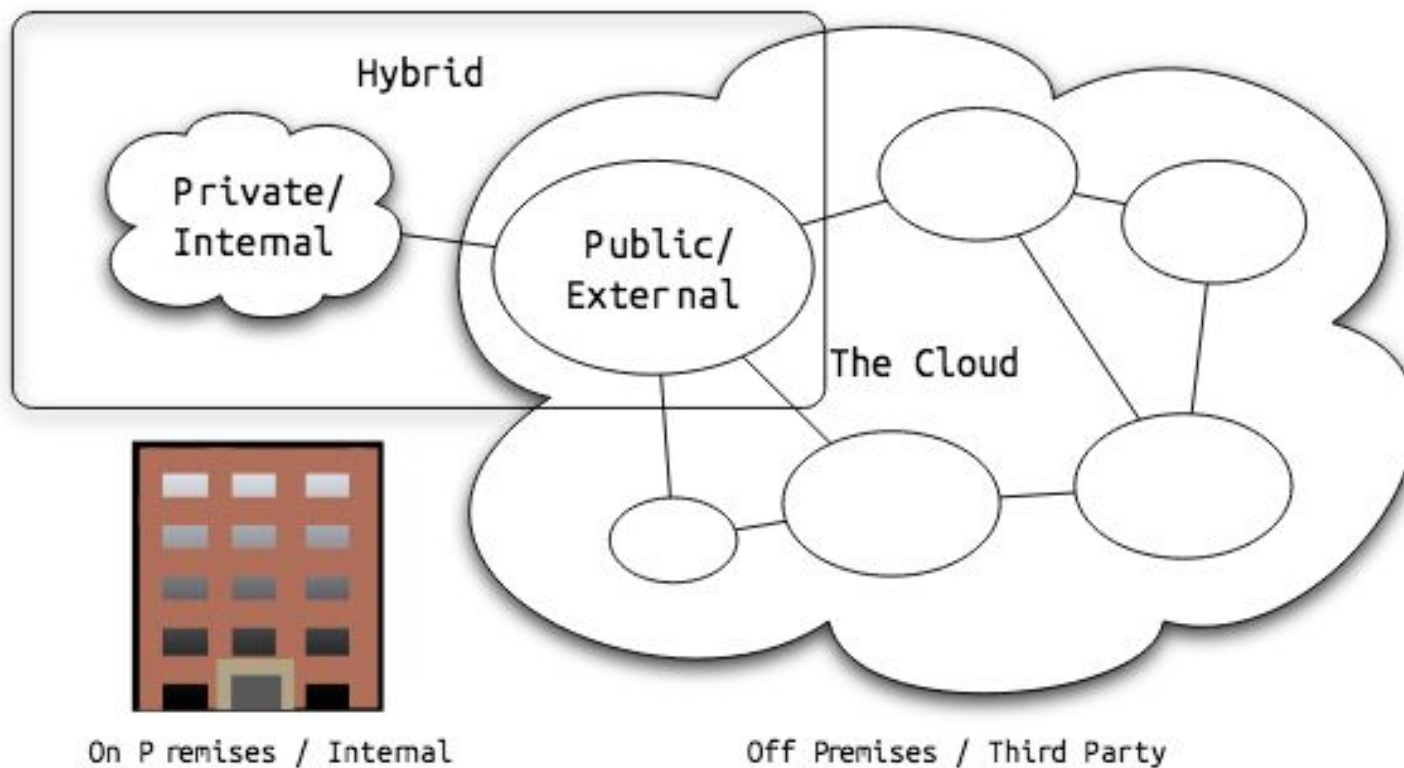
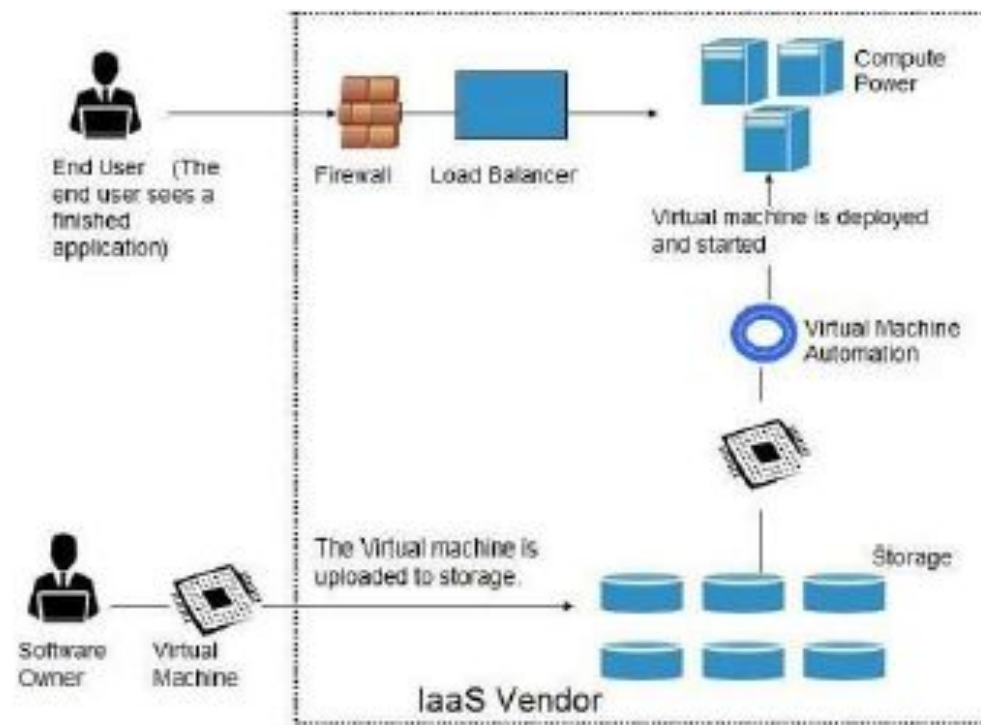


Recap

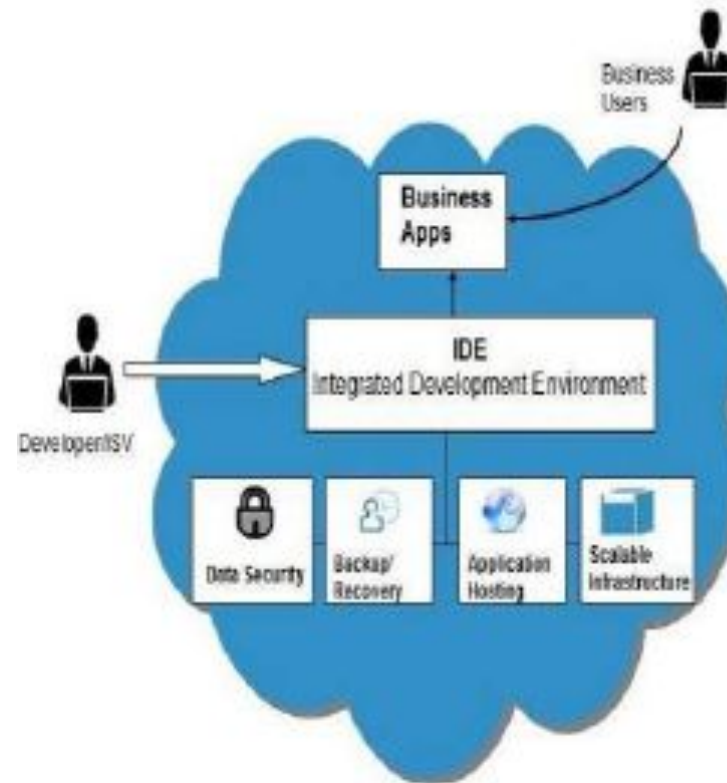


Cloud Computing T ypes

Recap – IaaS



Recap – PaaS



Recap – SaaS

Google's G Suite

Microsoft Office
365

Salesforce

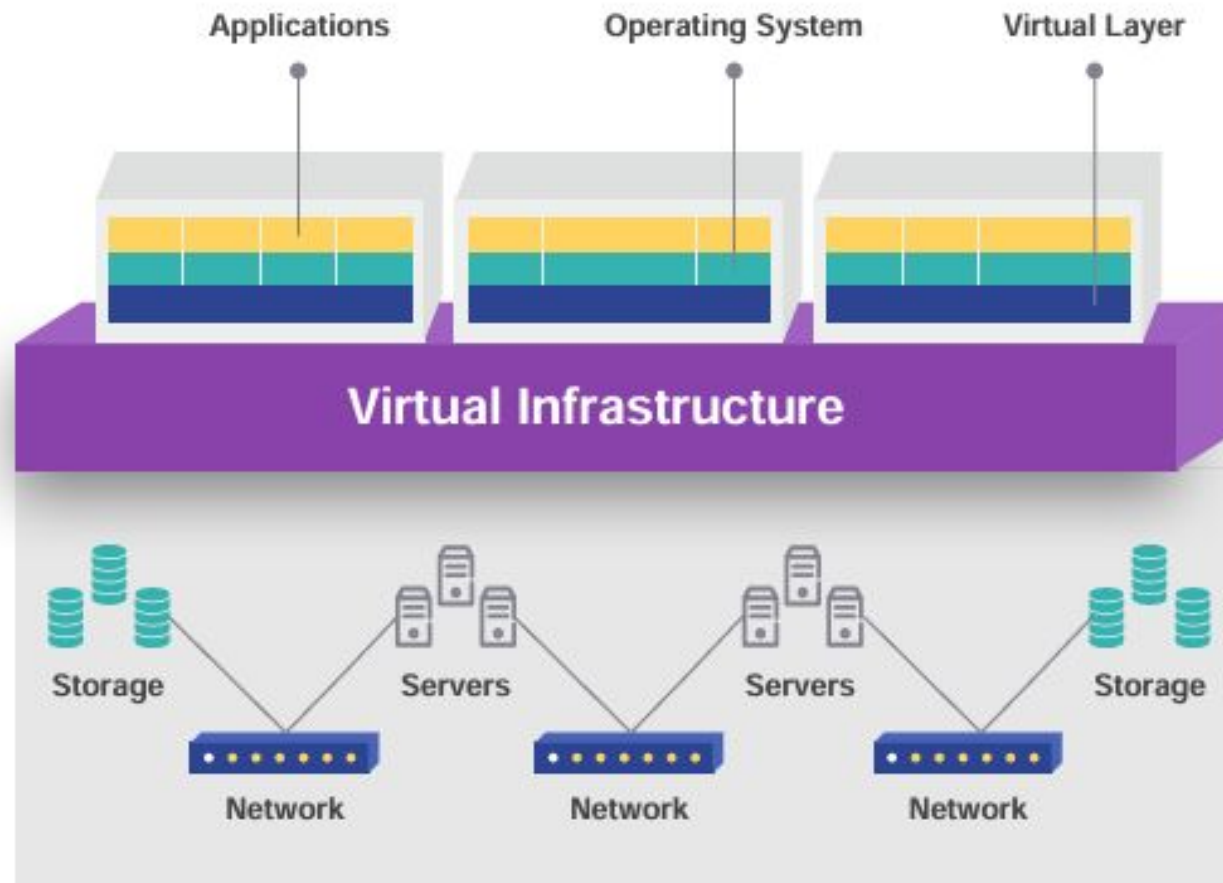


Virtualization

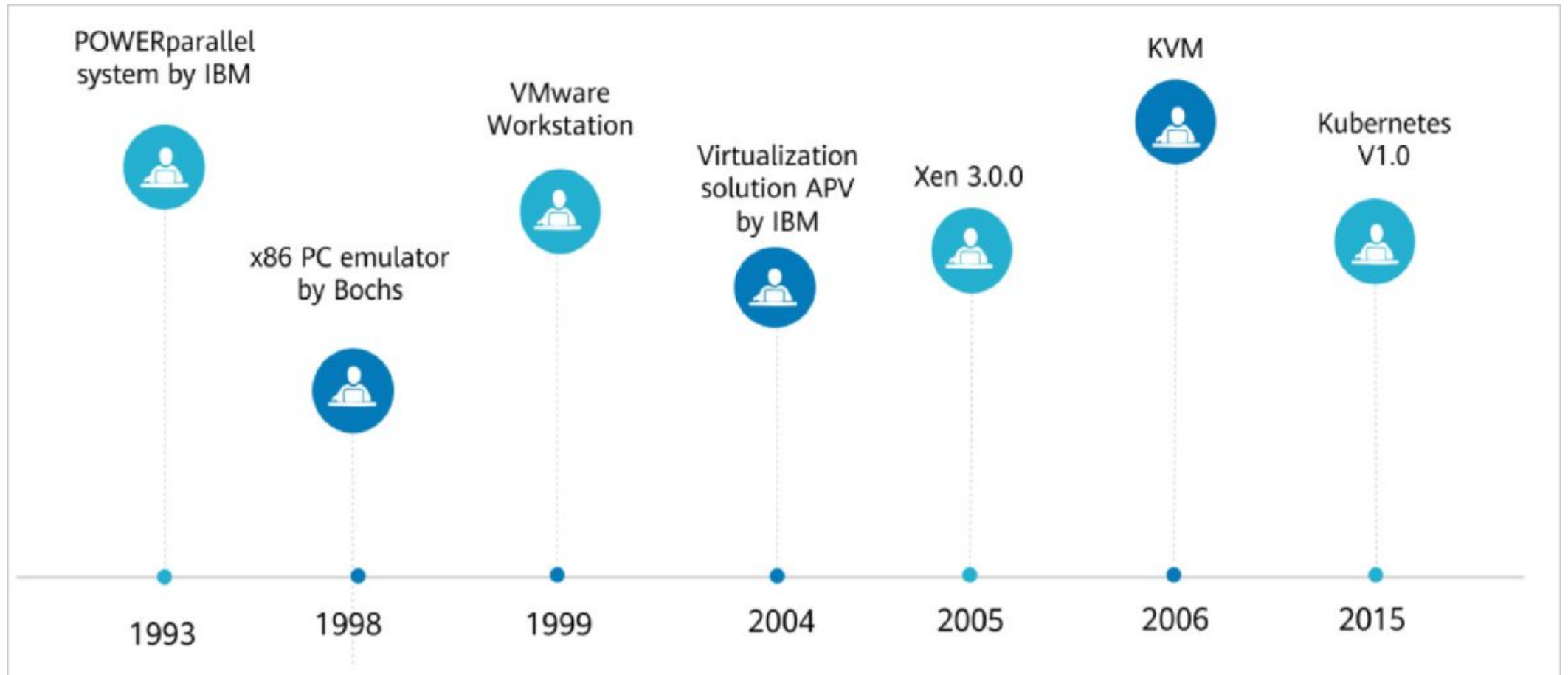
- Virtualization is technology that user can use to create virtual representations of servers, storage, networks, and other physical machines.
- Virtual software mimics the functions of physical hardware to run multiple virtual machines simultaneously on a single physical machine.
- The process of virtualization is a way to create multiple virtually simulated instances over the computer hardware to utilize user system's underlying resources fully.



Virtualization



Virtualization



Virtualization

- In 1993, IBM launched an upgradeable POWER parallel system, the first microprocessor-based supercomputer using RS/6000 technology.
- In 1998, Bochs released x86 PC emulator.
- In 1998, VMware was founded. In 1999, the company launched its first product, VMware Workstation, the commercial virtualization software that allows to run multiple operating systems on a single physical server. Since then, virtualization technology has been widely applied.
- In 1999, IBM first proposed the logical partitioning (LPAR) virtualization for the AS/400 system.
- In 2000, Citrix released XenDesktop, a desktop virtualization product.

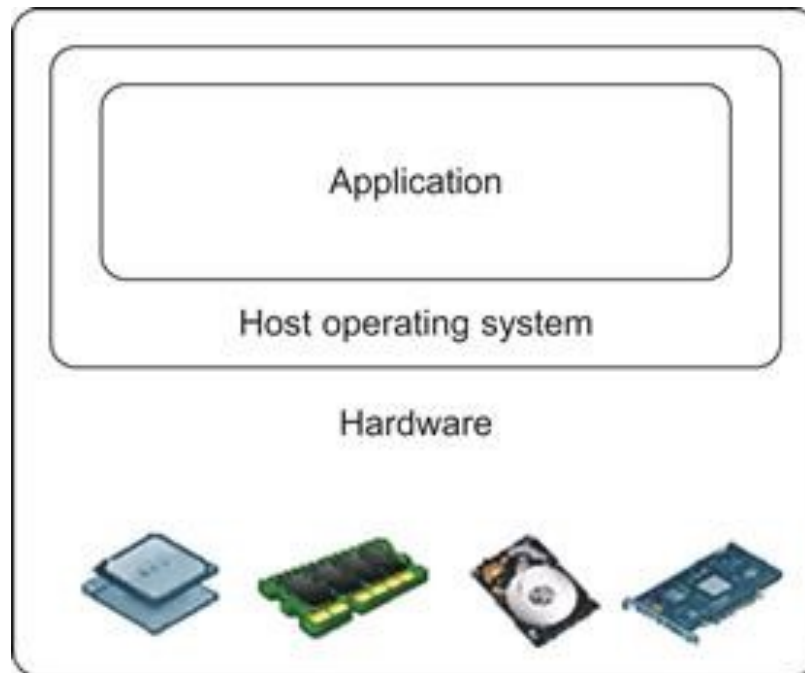
Virtualization

- In 2004, IBM released the virtualization solution Advanced Power Virtualization (APV), which supports resource sharing. This solution was later renamed PowerVM in 2008.
- In 2005, Xen 3.0.0 was released as the first hypervisor with Intel® VT-x support. Xen 3.0.0 can run on 32-bit servers.
- In 2006, Qumranet, an Israeli startup, officially announced Kernel-based Virtual Machine (KVM).
- 2006–present defines cloud computing and big data era.
- In 2007, InnoTek, a German company, developed VirtualBox.
- In 2008, Linux Container (LXC) 0.1.0 was released to provide lightweight virtualization.
- In 2010, Red Hat released RHEL 6.0, removing XEN and leaving KVM as the only bundled virtualization option.
- In 2015, Kubernetes v1.0 was released, opening the cloud native era.

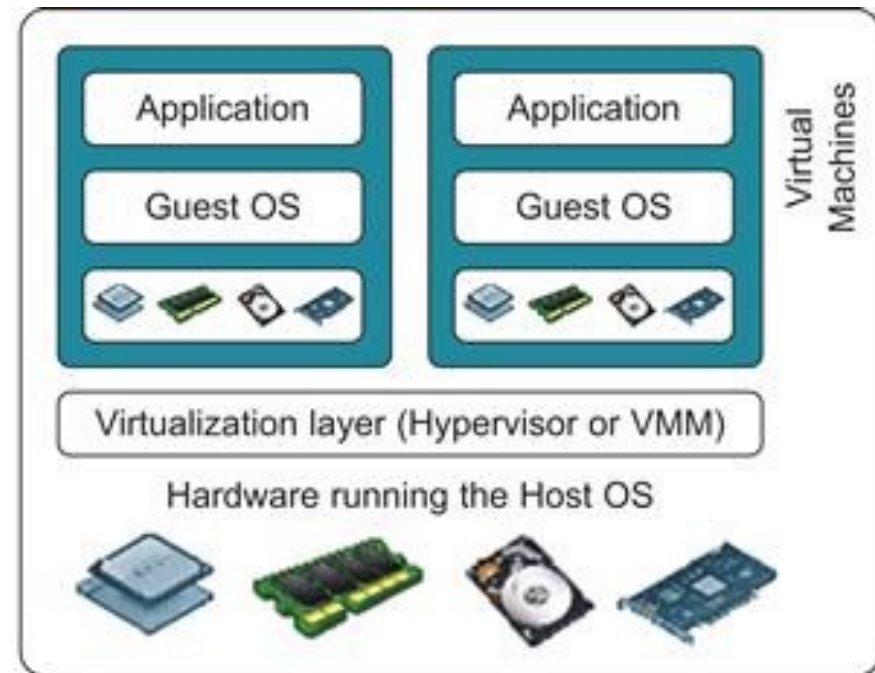
Need of Virtualization

1. Share same hardware among independent users- Degrees of Hardware parallelism increases.
2. Reduced Hardware footprint through consolidation- Eases management and energy usage.
3. Sandbox/migrate applications- Flexible allocation and utilization.
4. Decouple applications from underlying Hardware- Allows Hardware upgrades without impacting an OS image.

Need of Virtualization



(a) Traditional computer



(b) After virtualization

Need of Virtualization

- Virtualization is essential in cloud computing as it allows for the **efficient use of physical hardware resources by creating virtual instances of servers, storage devices, and networks**. This virtualization technology enables multiple virtual machines to run on a single physical server, optimizing resource utilization and reducing hardware costs.
- Virtualization in cloud computing provides a flexible and scalable environment for students to experiment with different configurations, operating systems, and applications. This flexibility allows for **easy setup of complex systems** and facilitates learning by doing.

Need of Virtualization

- Virtualization helps in creating isolated and secure environments for training programs, ensuring that each student has their own **virtual space to work on without interfering with others**. This isolation is crucial for maintaining the integrity of the training environment and protecting students' work.
- With virtualization in cloud computing, **students can access their virtual environments from anywhere with an internet connection, allowing for remote learning and collaboration**. This accessibility makes it convenient for students to continue their training programs from different locations and devices.

Current IT Scenarios - Virtualization

1. Cloud Computing

- **Example: Amazon Web Services (AWS)**
- AWS utilizes virtualization to provide scalable cloud services. Users can create and manage virtual servers (EC2 instances) that can be spun up or down based on demand. This flexibility allows businesses to only pay for what they use, optimizing costs.

2. Server Consolidation

- **Example: Data Centers**
- Many organizations have reduced their physical server count by using hypervisors like VMware vSphere or Microsoft Hyper-V. For instance, a company might consolidate 20 physical servers into 5 powerful servers running multiple VMs, significantly reducing hardware and maintenance costs.

Current IT Scenarios - Virtualization

3. DevOps Practices

Example: Continuous Integration/Continuous Deployment (CI/CD)

In a DevOps environment, tools like Jenkins can leverage virtualization to create isolated environments for each build. This allows developers to test their applications in various configurations quickly, ensuring compatibility and stability before deployment.

4. Enhanced Security

Example: Isolated Environments for Sensitive Applications

Organizations running financial or healthcare applications may deploy these on separate VMs, isolated from other systems. If one VM is compromised, the attacker cannot easily access other VMs, enhancing overall security.

Current IT Scenarios - Virtualization

5. Disaster Recovery Solutions

Example: VMware Site Recovery Manager

This tool automates the recovery of VMs in a disaster scenario. For example, if a company's primary data center goes down, it can quickly failover to a secondary site with minimal downtime, ensuring business continuity.

6. Hybrid Environments

Example: Microsoft Azure Stack

Companies can run Azure services on-premises alongside cloud solutions, allowing for a hybrid cloud strategy. For instance, sensitive data can be kept on-premises while leveraging Azure for analytics, providing flexibility and compliance.

Current IT Scenarios - Virtualization

7. Resource Management

Example: VMware vRealize Operations

This tool helps monitor and manage resources in virtual environments. It provides insights into resource utilization, helping organizations optimize performance. For example, a company can identify underused VMs and consolidate workloads to improve efficiency.

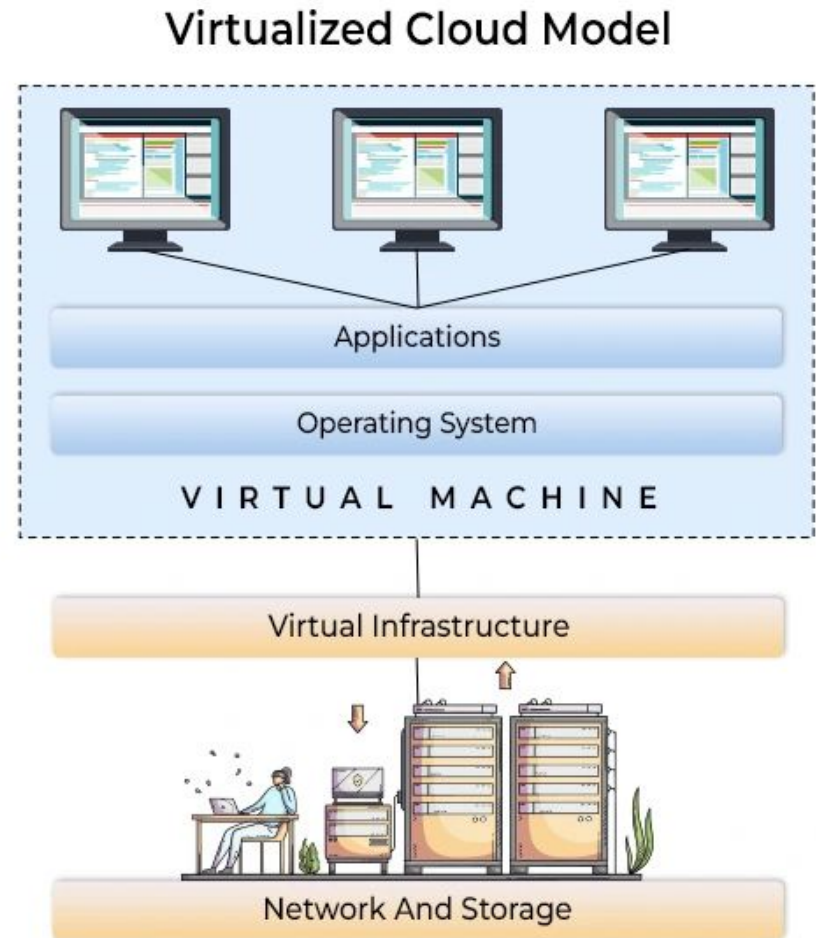
8. IoT Integration

Example: Edge Computing with Virtualization

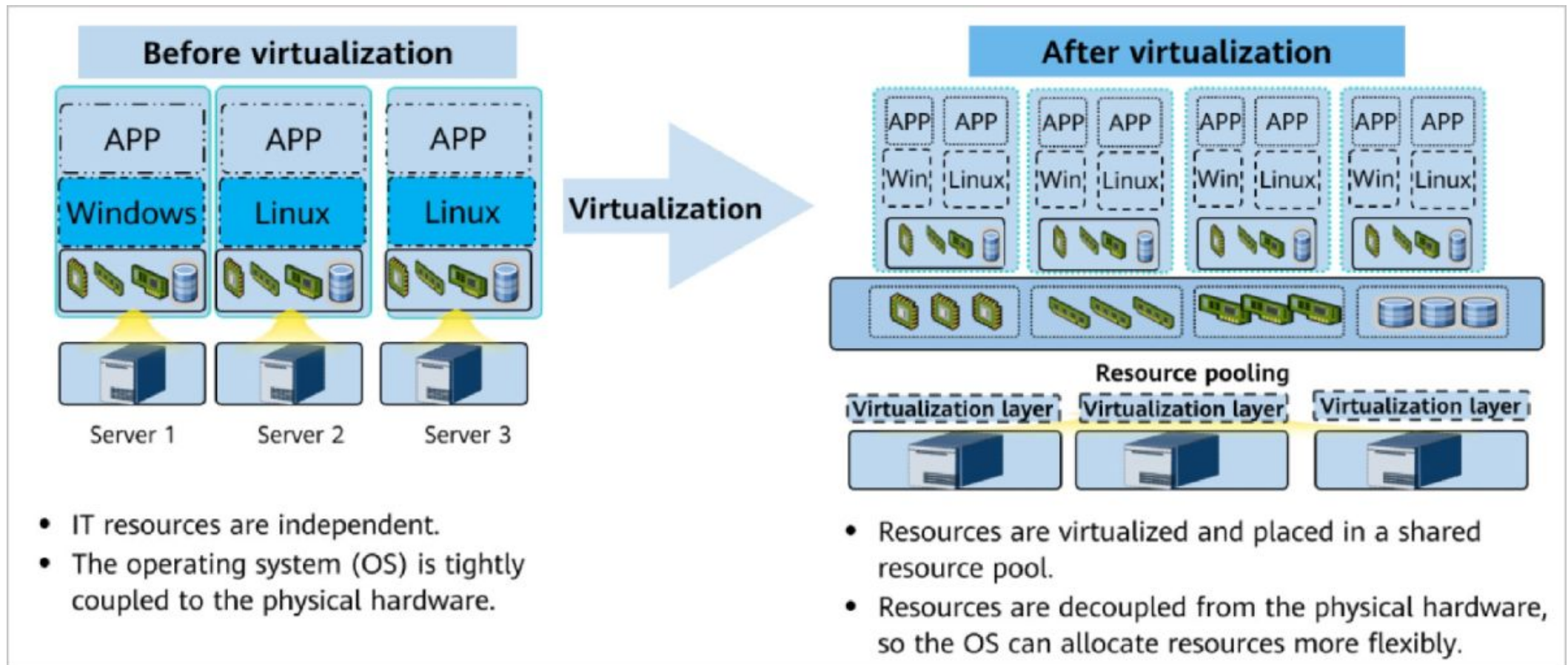
In IoT scenarios, edge devices can utilize virtualization to run applications closer to the data source. For instance, a smart manufacturing facility might deploy VMs on edge servers to process data from sensors in real-time, reducing latency and improving responsiveness.

Virtualization vs. Non- Virtualization

- The difference between virtualization vs. cloud computing is that virtualization is a technology that enables **a single server to provide capacity for many**.
- In contrast, cloud computing allows **different computers to connect with shared resources via the internet**.



Virtualization vs. Non- Virtualization



Virtualization vs. Non- Virtualization

Virtualization and non-virtualization represent two distinct approaches to managing computing resources. Let's explore the differences:

Virtualization:

Definition: Virtualization involves creating virtual instances of operating systems, servers, storage, or network resources. These virtual machines (VMs) run on a physical host but operate independently.

Advantages:

- 1. Resource Utilization:** Virtualization allows efficient use of hardware by running multiple VMs on a single physical server.
- 2. Isolation:** VMs are isolated from each other, enhancing security and stability.
- 3. Flexibility:** Easily scale resources up or down as needed.

Examples: VMware, Hyper-V, KVM.

Virtualization vs. Non- Virtualization

Non-Virtualization:

Definition: In a non-virtualized environment, applications run directly on physical hardware without an abstraction layer.

Advantages:

- 1. Performance:** No overhead from virtualization, leading to better performance.
- 2. Simplicity:** Fewer layers mean simpler management.
- 3. Specific Use Cases:** Some workloads (e.g., high-performance computing) benefit from direct hardware access.

Examples: Traditional bare-metal servers, dedicated appliances.

Understanding Virtualization

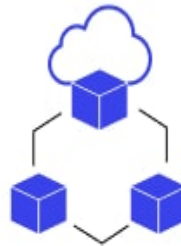
- **Virtual machine manager (VMM):** A VMM, also called the hypervisor, is software that creates and manages virtual machines. It is a specialized operating system that uses minimal hardware resources so that memory and processing are available for the VMs it creates.
- **Virtual machine (VM):** A VM is a machine created by a hypervisor/VMM that runs like any other computer. It usually needs an operating system installed on it to become functional. A VM uses the VMM/hypervisor for access to memory, CPU, network, video, and other resources.

Understanding Virtualization Technologies

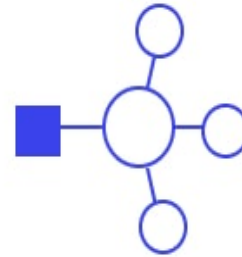
Types of Virtualization



Desktop



Data



Network



Storage



Server



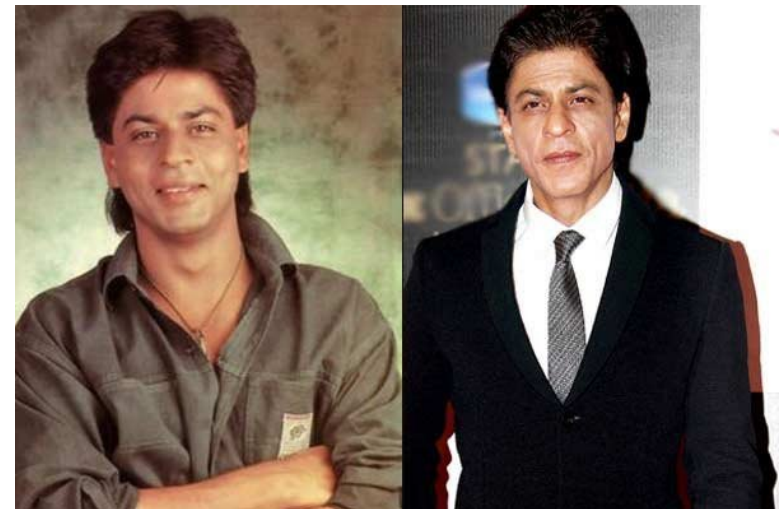
Application



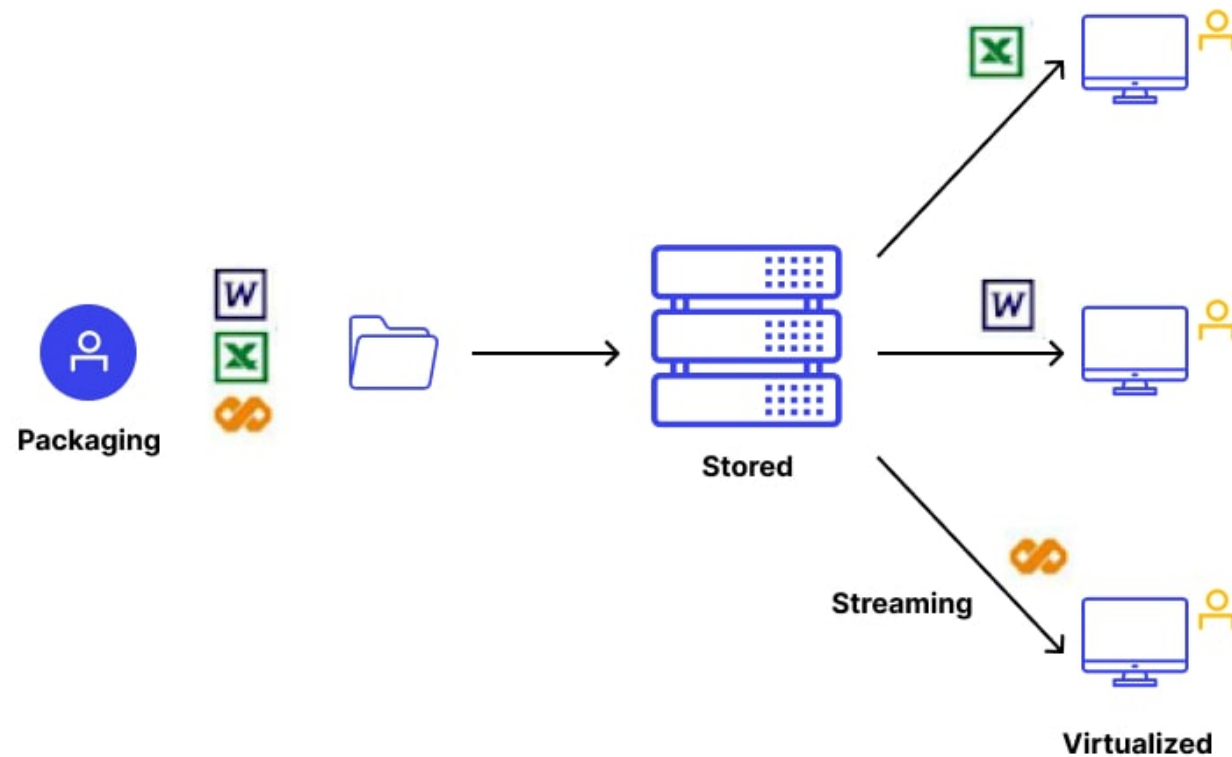
Cloud

Application Virtualization

- Application virtualization helps a user to have remote access to an application from a server.
- The server stores all personal information and other characteristics of the application but can still run on a local workstation through the internet.
- **An example of this would be a user who needs to run two different versions of the same software.** Technologies that use application virtualization are hosted applications and packaged applications.



Application Virtualization



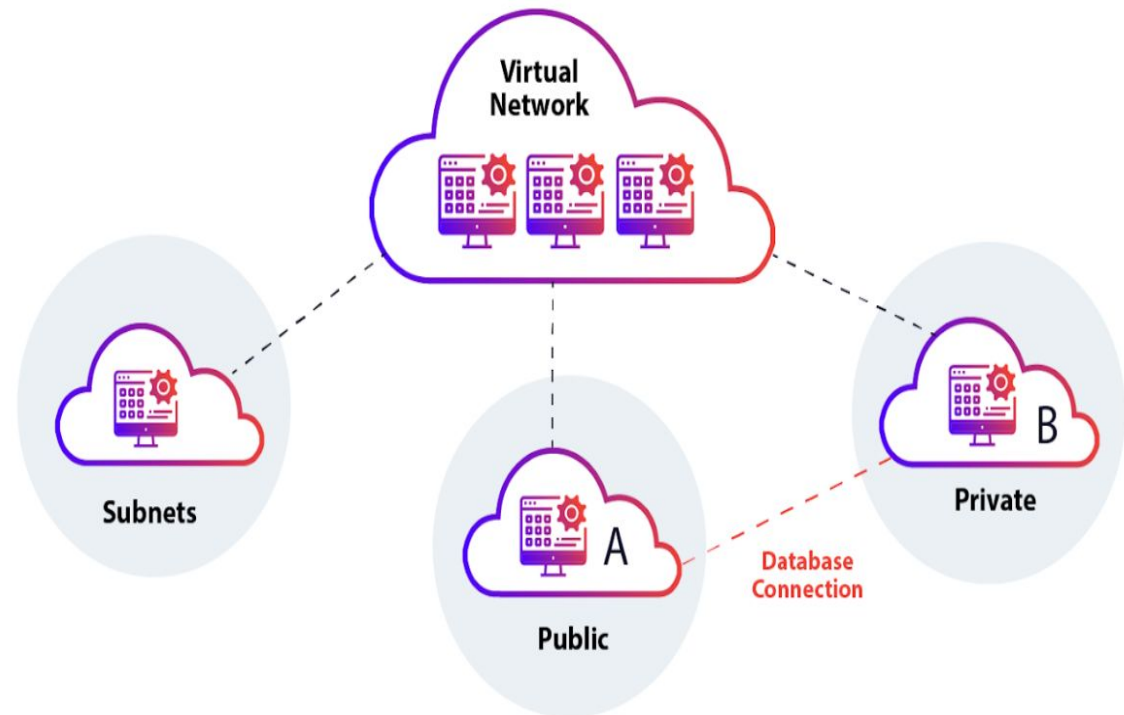
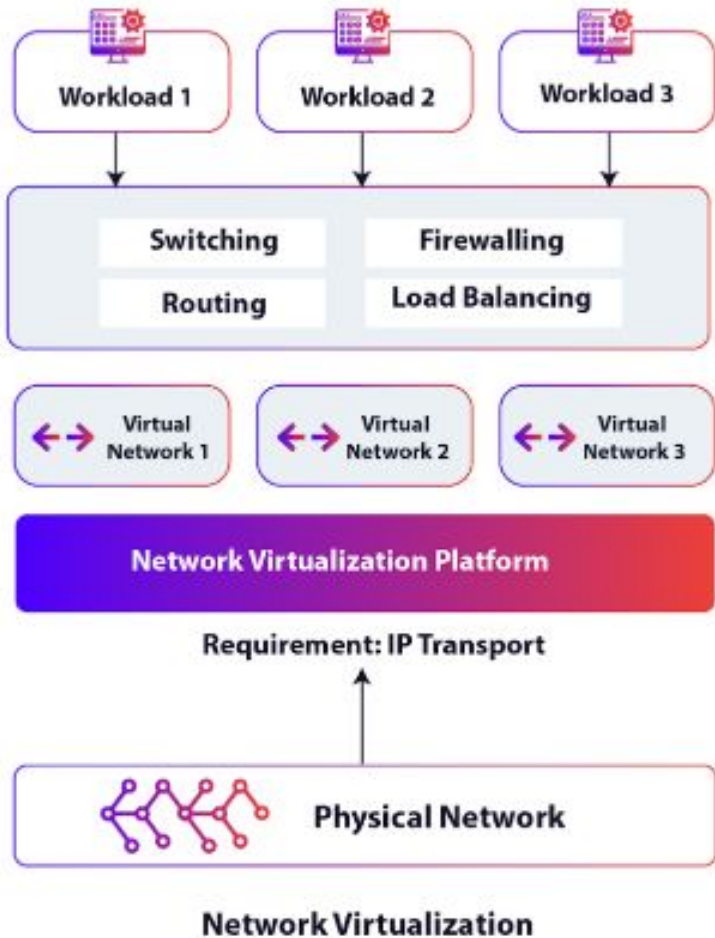
Network Virtualization

Network Virtualization:

- Network virtualization provides a facility to create and provision virtual networks—logical switches, routers, firewalls, load balancers, VPN, and workload security within days or weeks.
- It is the ability to run multiple virtual networks with separate control and data plan.
- **It co-exists on top of one physical network.** It can be managed by potentially confidential individual parties.
 - Network Virtualization increase agility
 - Network Virtualization increase efficiency
 - Reduces the operational costs
 - Able to maintain high standards for security

Network Virtualization

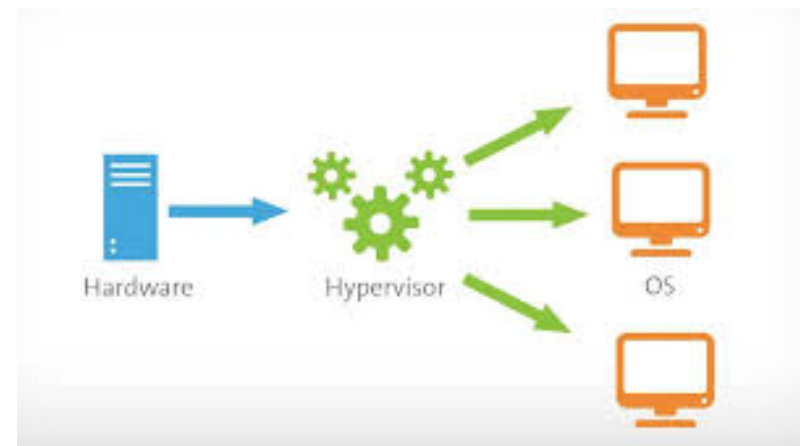
Network Virtualization:



Server Virtualization

Server Virtualization:

- Server Virtualization is the process of **dividing a physical server into several virtual servers, called virtual private servers**. Each virtual private server can run independently.
- The concept of Server Virtualization widely used in the IT infrastructure to minimizes the costs by increasing the utilization of existing resources



Server Virtualization – Hypervisor

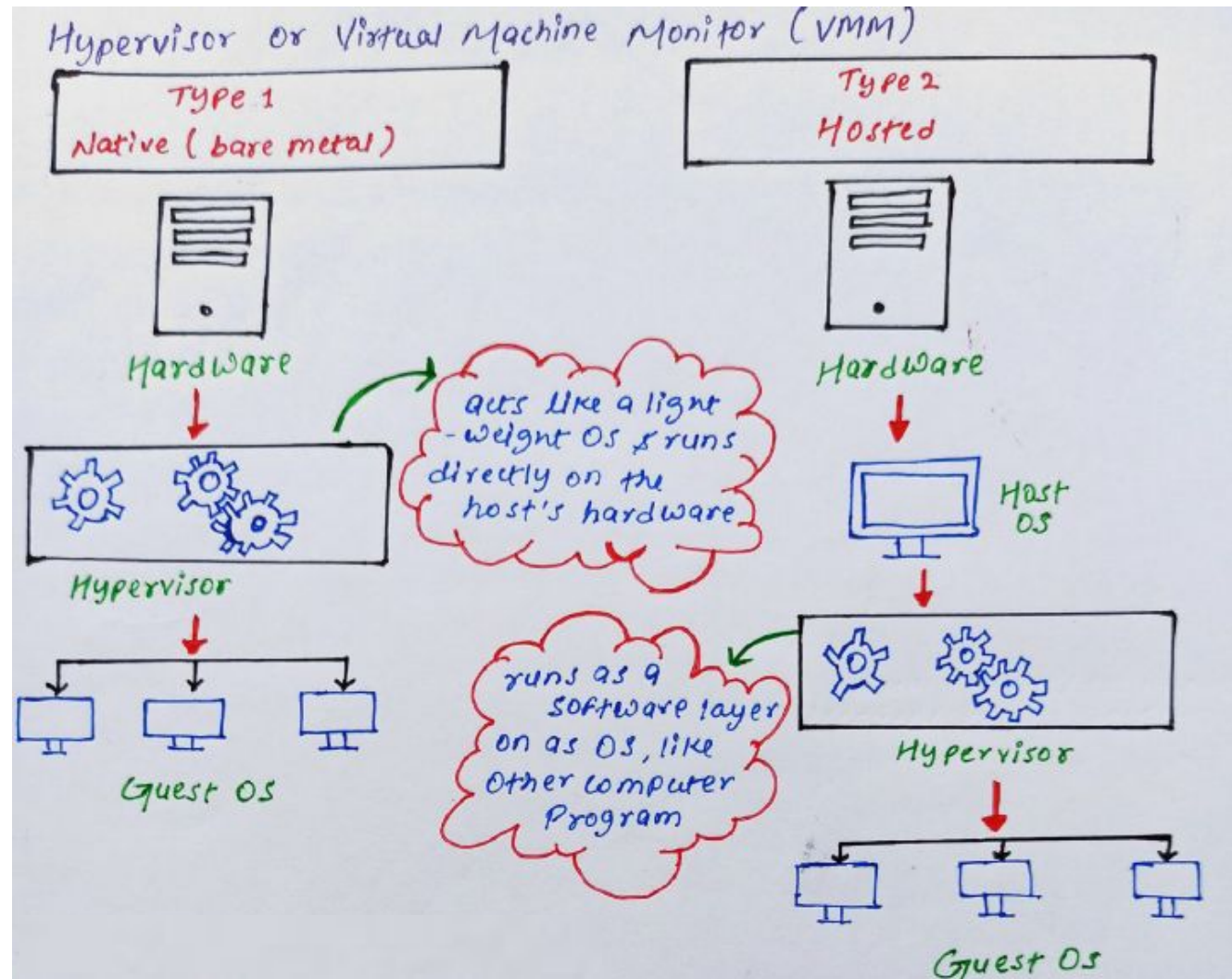
Types of Server Virtualization:

1. Hypervisor

- In the Server Virtualization, Hypervisor plays an important role. It is a layer between the operating system (OS) and hardware. There are two types of hypervisors.
 - Type 1 hypervisor (also known as bare metal or native hypervisors)
 - Type 2 hypervisor (also known as hosted or Embedded hypervisors)
- The hypervisor is mainly used to perform various tasks such as allocate physical hardware resources (CPU, RAM, etc.) to several smaller independent virtual machines, called "guest" on the host machine.

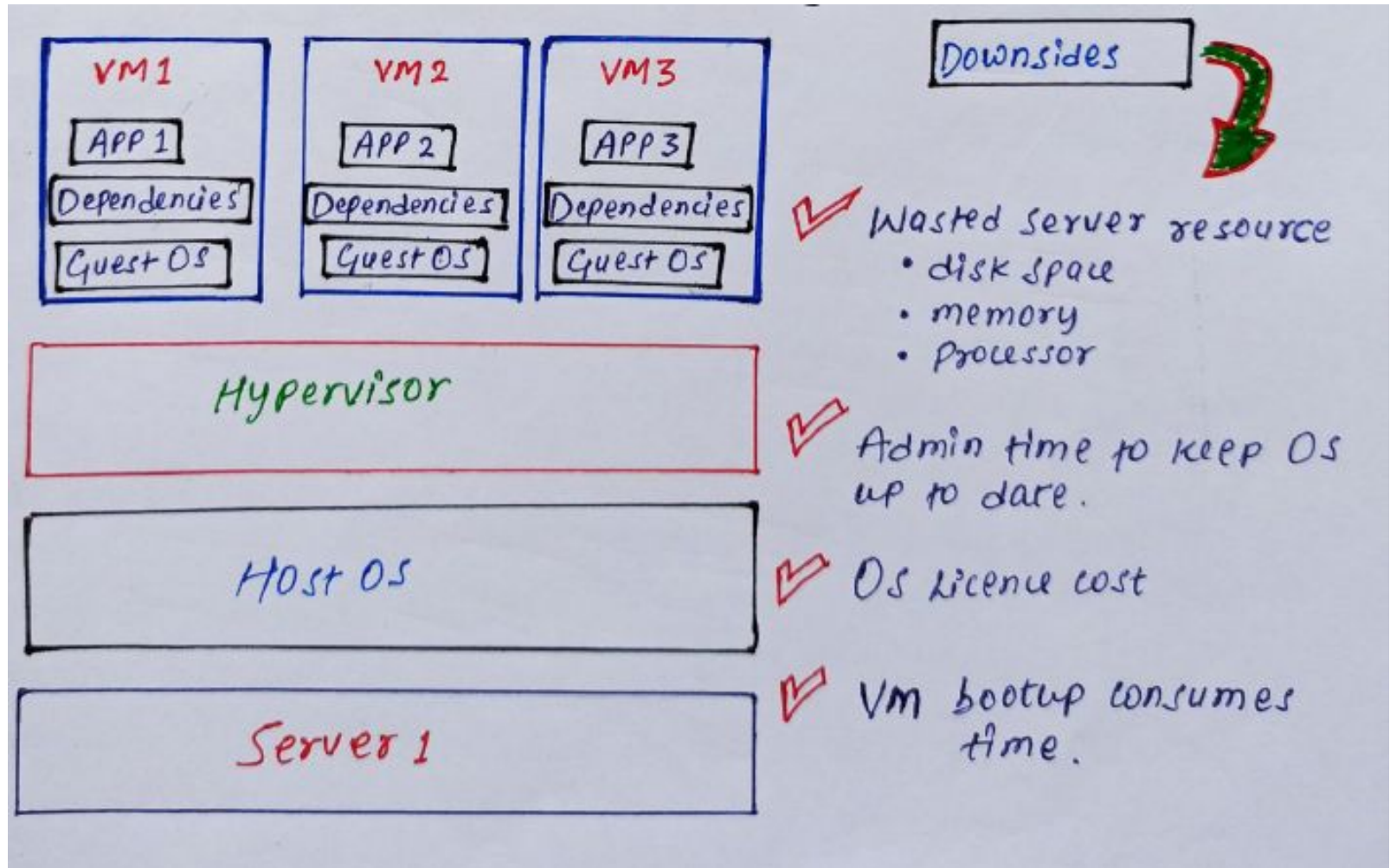
Server Virtualization – Hypervisor

Hypervisor



Server Virtualization – Hypervisor

Hypervisor



Server Virtualization – Full

Types of Server Virtualization:

2. Full Virtualization

- Full Virtualization **uses a hypervisor to directly communicate with the CPU and physical server.**
- It provides the best isolation and security mechanism to the virtual machines.
- The biggest disadvantage of using hypervisor in full virtualization is that a hypervisor has its own processing needs, so it can slow down the application and server performance.
- VMWare ESX server is the best example of full virtualization.

Server Virtualization – Full

- Full virtualization allows multiple guest operating system to execute on a host operating system independently.
- The guest operating systems do not concern about the presence of a hypervisor. Therefore, each virtual machine and its guest operating system operate as independent computers.
- Full Virtualization is slow.
- Microsoft and Parallels systems are some examples of Full virtualization

Server Virtualization – Full

How Full Virtualization Works?

1. Hypervisor Layer:

Full virtualization relies on a hypervisor, also known as a **Virtual Machine Monitor (VMM)**, which sits between the physical hardware and guest operating systems. The hypervisor manages and controls the allocation of physical resources to virtual machines.

2. Hardware Virtualization:

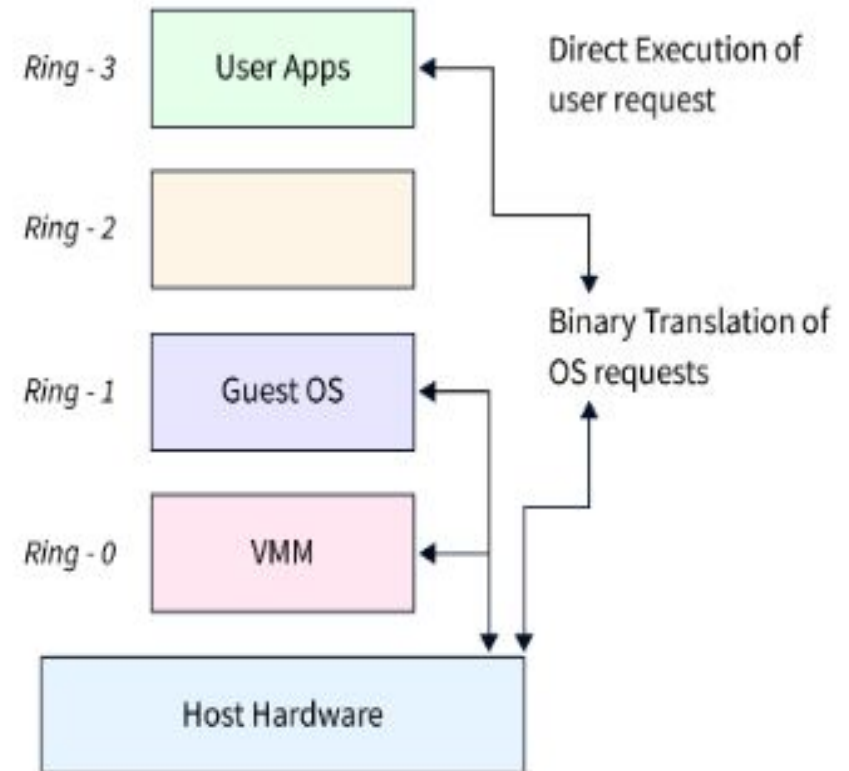
Full virtualization uses hardware-assisted virtualization technologies like **Intel VT-x** or **AMD-V** to enhance performance. These technologies allow the hypervisor to run guest OSes directly on the physical CPU without significant performance overhead.

3. Isolation:

VMs created through full virtualization are completely isolated from each other. Each VM runs its own instance of the guest operating system, which cannot interfere with other VMs.

4. Examples:

Popular hypervisors for full virtualization include **VMware vSphere/ESXi**, **Microsoft Hyper-V**, and **Oracle VirtualBox**.



Para Virtualization

Para Virtualization:

- Para Virtualization is quite similar to the Full Virtualization. The advantage of using this virtualization is that it is easier to use, Enhanced performance, and does not require emulation overhead.
- Xen primarily and UML use the Para Virtualization.
- The difference between full and pare virtualization is that, **in para virtualization hypervisor does not need too much processing power to manage the OS**

Para Virtualization

- Paravirtualization uses hypercalls for operations to handle instructions at compile time.
- Guest OS is not completely isolated but it is partially isolated by the virtual machine from the virtualization layer and hardware.
- Paravirtualization is faster in operation and is more secure.
- It is less portable and compatible.
- VMware and Xen are some examples of paravirtualization.

Para Virtualization

Para Virtualization

1. Hypervisor Layer:

Similar to full virtualization, paravirtualization also employs a hypervisor, but here, the guest operating systems are aware of it. The hypervisor provides a set of APIs that guest OSes must use to communicate with the underlying hardware.

2. Guest OS Modifications:

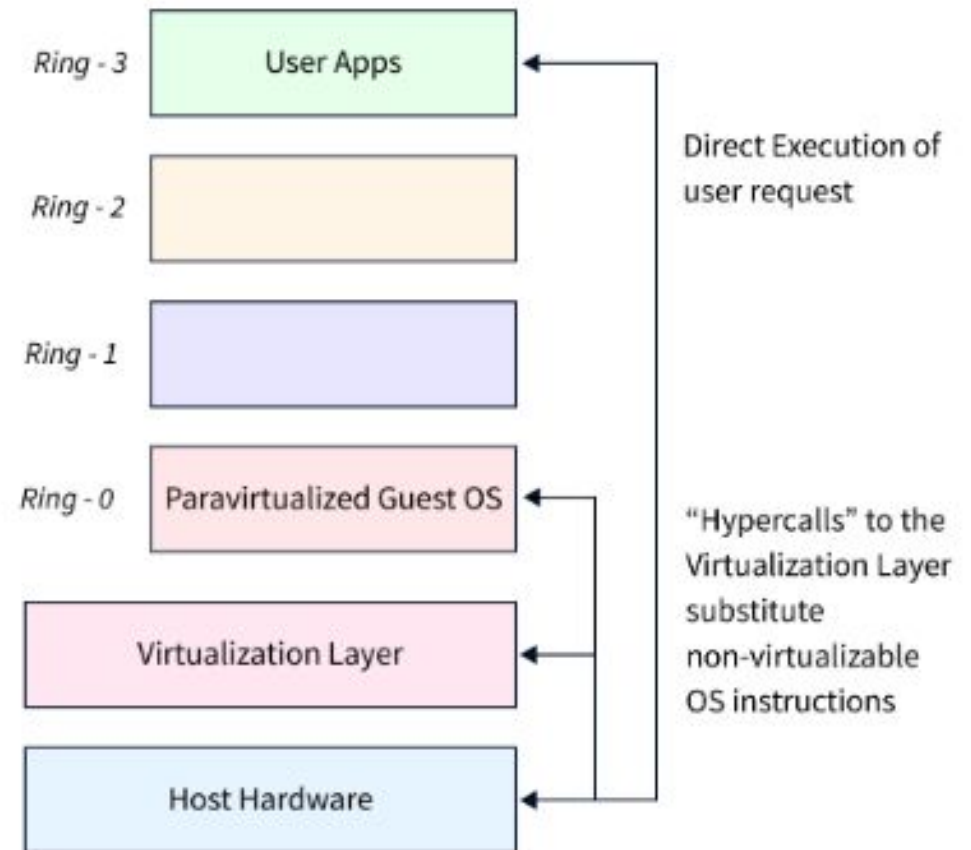
Guest operating systems must be modified to replace certain hardware-related instructions with hypercalls, which are calls to the hypervisor. These hypercalls allow the guest OS to request services from the hypervisor, such as memory management or CPU scheduling.

3. Performance Benefits:

Since para virtualization avoids the overhead of emulating complete hardware, it often offers better performance than full virtualization. Guest OSes can communicate more directly with the hypervisor, resulting in improved efficiency.

4. Examples:

Xen is a widely-used hypervisor that supports para virtualization. It is known for its performance and scalability in virtualized environments.



Full Virtualization vs Para Virtualization

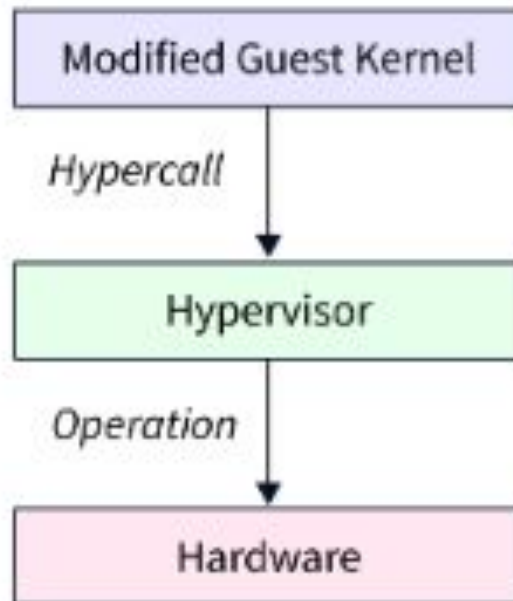
Aspect	Full Virtualization	Para Virtualization
Guest OS Modification	Not required; runs unmodified	Requires modifications to use hypercalls
Performance	Slightly lower due to emulation	Better performance due to direct interaction
Isolation	Strong isolation between VMs	Isolation with awareness of other VMs
Guest OS Flexibility	Supports various OS types	Works best with compatible OSes
Hardware Compatibility	Compatible with most hardware	Requires hardware support for para virtualization

Full Virtualization vs Para Virtualization

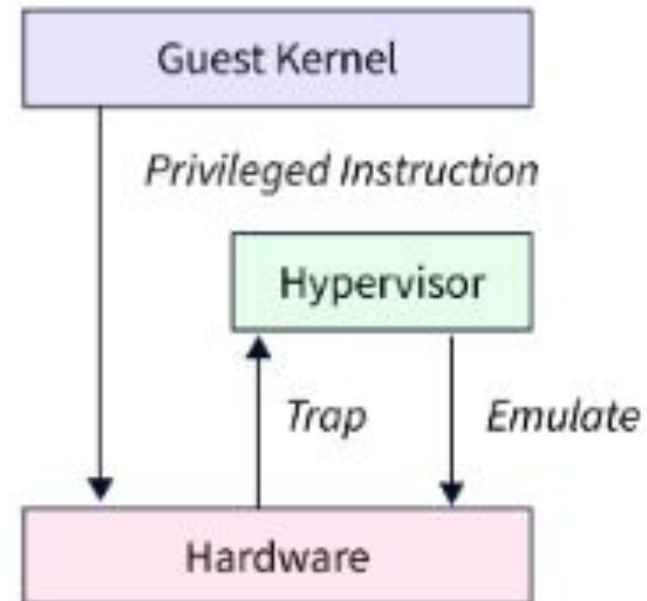
Aspect	Full Virtualization	Para Virtualization
Hypervisor Layer	Hypervisor manages virtual hardware independently	Hypervisor provides APIs for communication
Interaction with Hardware	Emulates complete hardware	Uses hypercalls to request hardware services
Examples	VMware, Hyper-V, VirtualBox	Xen, KVM, QEMU
Resource Overhead	Slightly higher resource overhead	Lower resource overhead for the hypervisor
Use Cases	Mixed OS environments, strong isolation needed	Performance-critical applications, homogeneous OS environment

Full Virtualization vs Para Virtualization

Para-virtualization



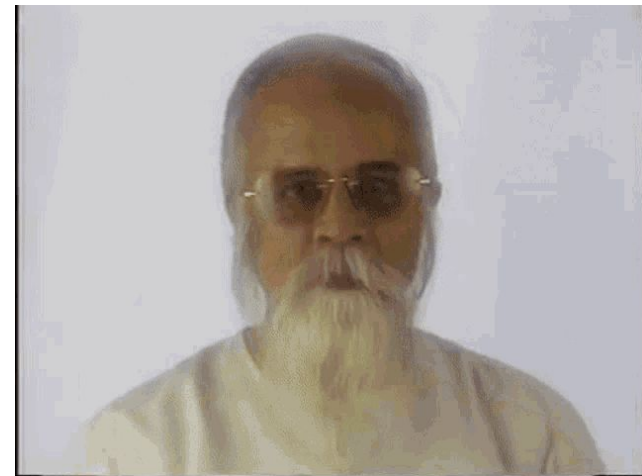
"Classical" Full Virtualization



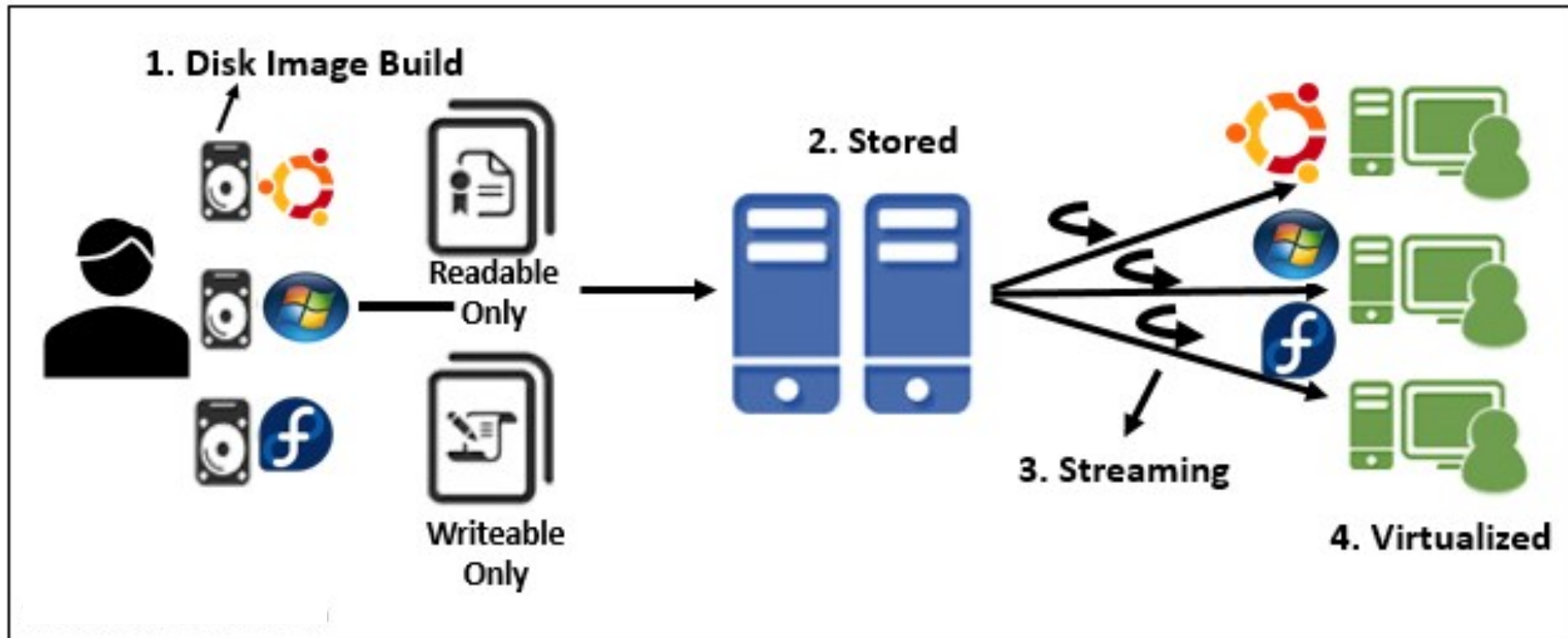
Operating System Virtualization

Operating System Virtualization:

- Operating system virtualizations includes a modified form than a normal operating system so that different users can operate its end use different applications. This whole process shall perform on a single computer at a time.
- Linux OS Virtualization and Windows OS Virtualization are the types of Operating System virtualization. FreeVPS, OpenVZ, and Linux Vserver are some examples of System-Level Virtualization..



Operating System Virtualization



Operating System Virtualization

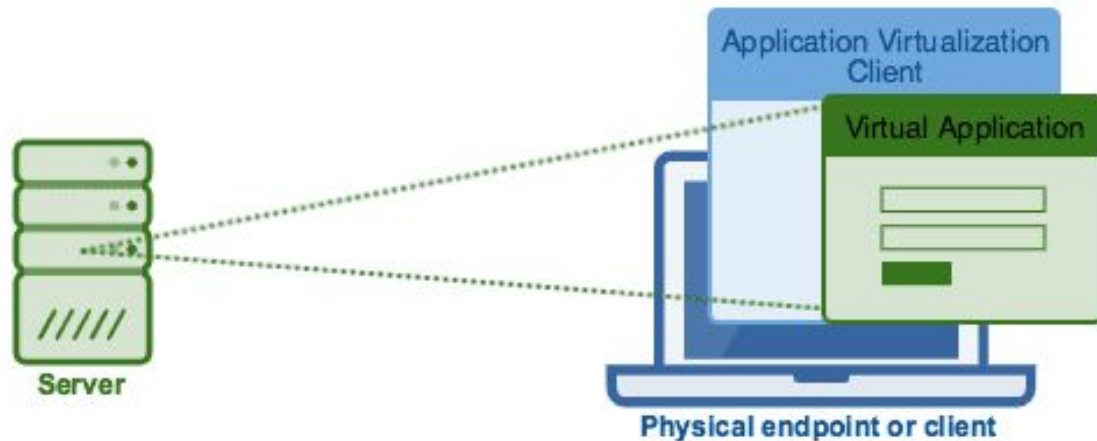
Operating System Virtualization:

- In operating system virtualizations when the application does not interfere with another one even though they are functioning in the same computer.
- The kernel of an operating system allows more than one isolated user-space instance to exist. These instances call as software containers, which are virtualizations engines.

Client Virtualization

Client Virtualization:

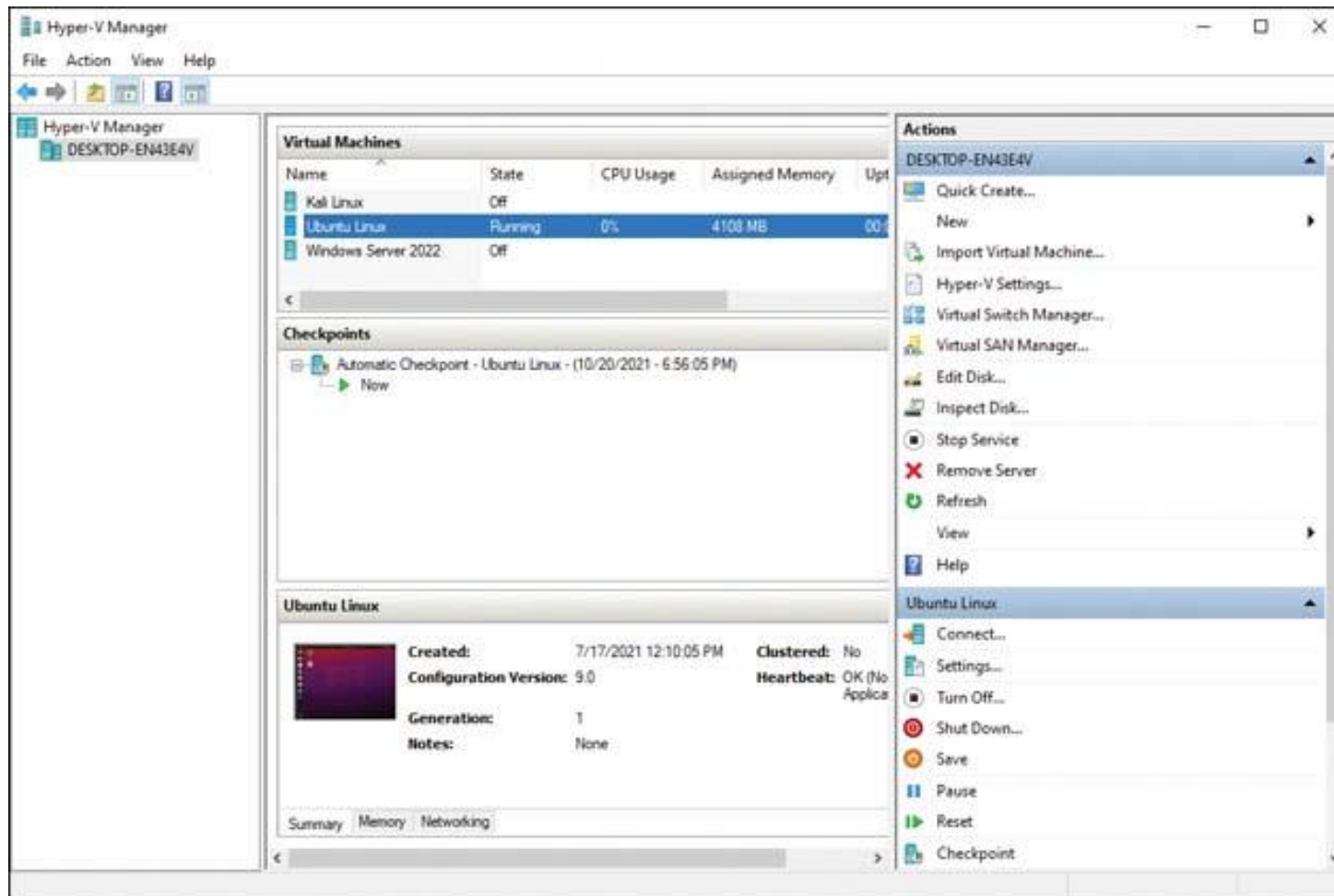
- Client virtualization is a virtual machine (VM) environment in the user's machine.
- Also called "endpoint virtualization," the user's computer hosts multiple VMs, each of which contains an operating system and set of applications.
- Similar like Desktop Virtualization



Client Virtualization



Client Virtualization



Client Virtualization

Advantages

- Enabling users to run applications that aren't native to their device's operating system, such as running Microsoft Word or Excel on a Chromebook or Linux laptop
- Repurposing legacy devices and enabling them to run remote applications
- Enhancing security by storing data on the server instead of the client
- Simplifying administration by centralizing application management and maintenance

Disadvantages

- Dependency to the client/applications machine

Application Streaming

- Application streaming is a related concept to application virtualization, where applications are run directly from a virtual machine on a central server that is completely separate from the local system.
- An application is packaged and stored on a streaming server.
- Packaging or sequencing produces an image of the application in a way that orders delivery or predicatively optimizes delivery to the client.
- **Virtual Desktop Infrastructure (VDI) is a software tool that allows users to access enterprise computer systems from almost any device, such as a tablet, smartphone, or personal computer**

Application Streaming

- App streaming is a process where end-point devices access virtualized apps from a remote server setup. The apps delivered are virtualized on a server.
- Virtual desktops are VDI-based desktops that are equipped with specific configurations and operating systems. They have dedicated resources, applications, and user settings.
- App streaming provides stand-alone applications, while virtual desktops offer dedicated desktop environments. Both have their own set of pros and cons. Companies must first define their requirements and then select between an app streaming solution and virtual desktop accordingly.

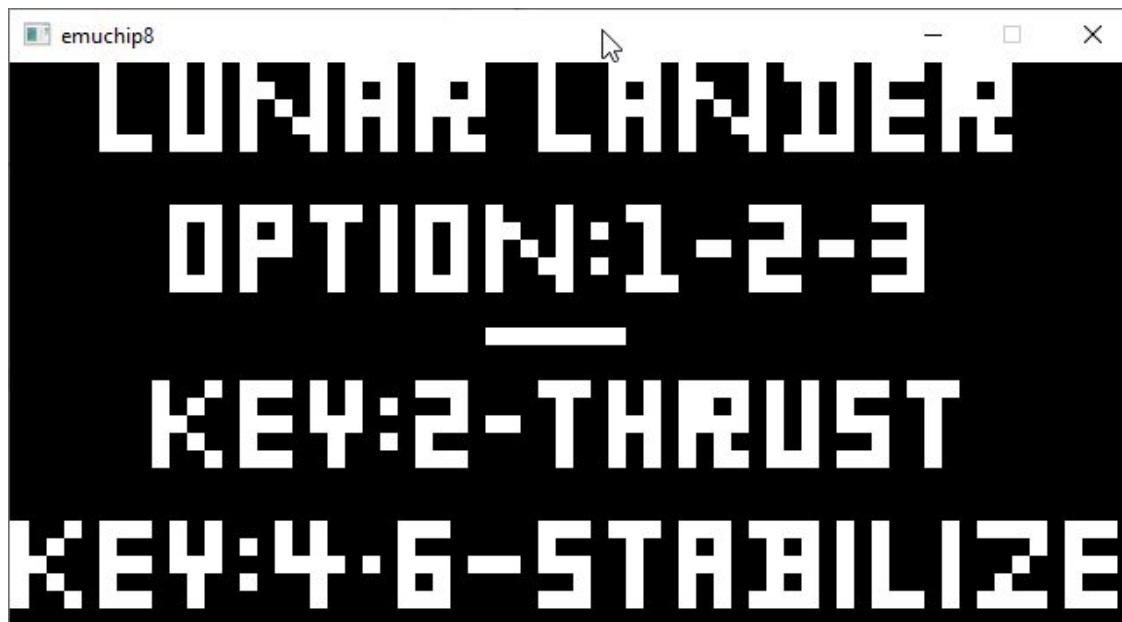
Hardware Emulation

- Hardware emulation is the use of one hardware device to mimic the function of another hardware device.
- A hardware emulator simulates the workings of an entirely different hardware platform than the one it runs on. Hardware emulation is often used to debug and verify systems under design.

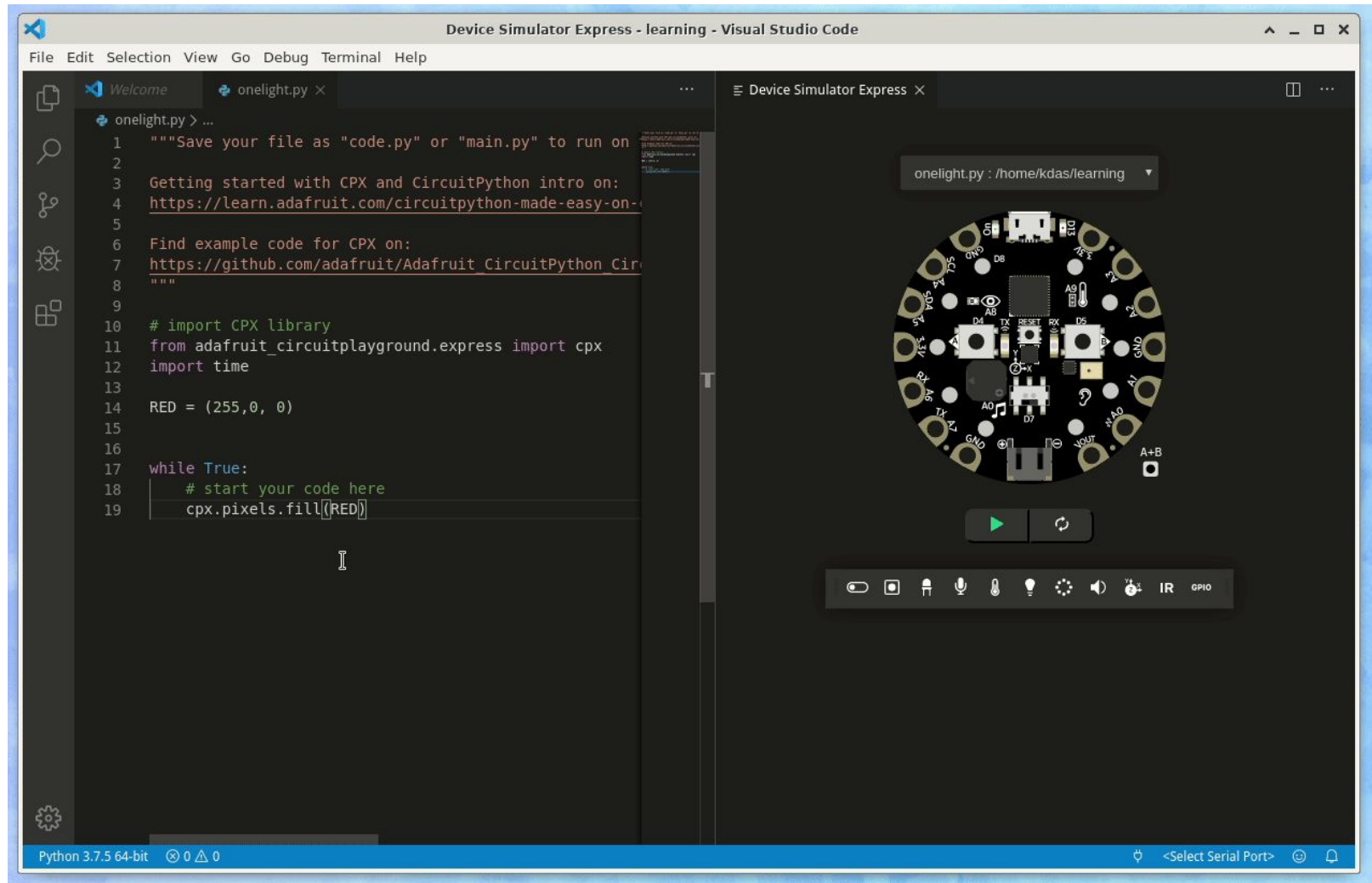


Hardware Emulation

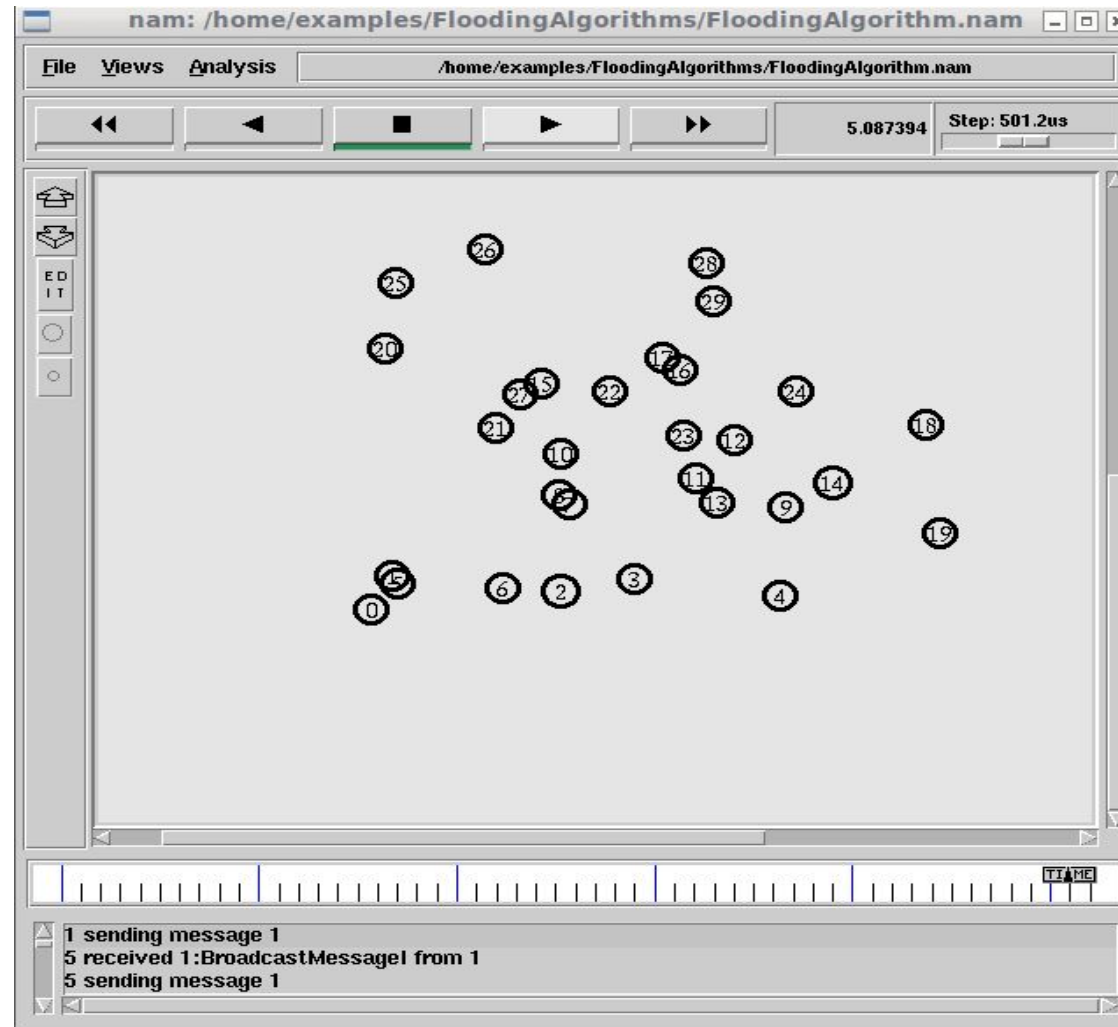
- IT administrators can use hardware emulation to run an unsupported operating system (OS) on a virtual machine (VM).
- In such a scenario, the VM does not have direct access to server hardware. Instead, an emulation layer directs traffic between physical and virtual hardware. This is less efficient than paravirtualization, in which the interface for a VM can differ somewhat from that of the underlying hardware.



Simulation



Simulation

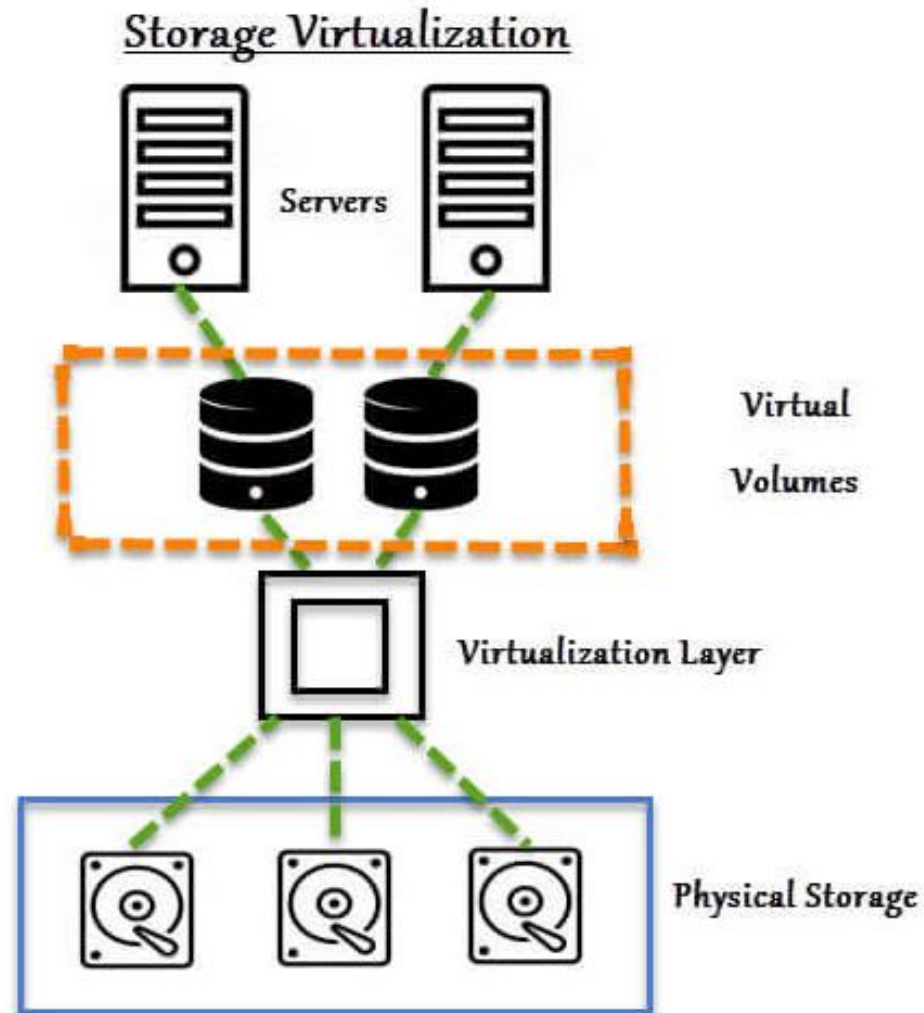


Storage Virtualization

- Storage virtualization in Cloud Computing is the sharing of physical storage into multiple storage devices which further appears to be a single storage device.
- It can be also called as a group of an available storage device which simply manages from a central console.



Storage Virtualization



Why Storage Virtualization

Perform Task

- The appliances of storage virtualization are responsible for several tasks such as heterogeneous replication and federation. These devices lineup in front of arrays and create a common interface for the host.

WAN Management

- It does not send multiple copies of the similar data over WAN. The WAN accelerator use to cache the data and send it LAN speed without changing the performance of WAN.

Why Storage Virtualization

Disaster Recovery

- Storage virtualization in Cloud Computing can increase disk utilization and is flexible. This ameliorates disaster recovery and the continuity of the business.

Storage Tiering

- Storage tiering is a technique which monitors and selects the most commonly used data and put it on its highest performing storage pool. The least used data is put in the weakest performance storage pool.

Why Storage Virtualization



Types of Storage Virtualization

Two types of storage virtualization, which are:

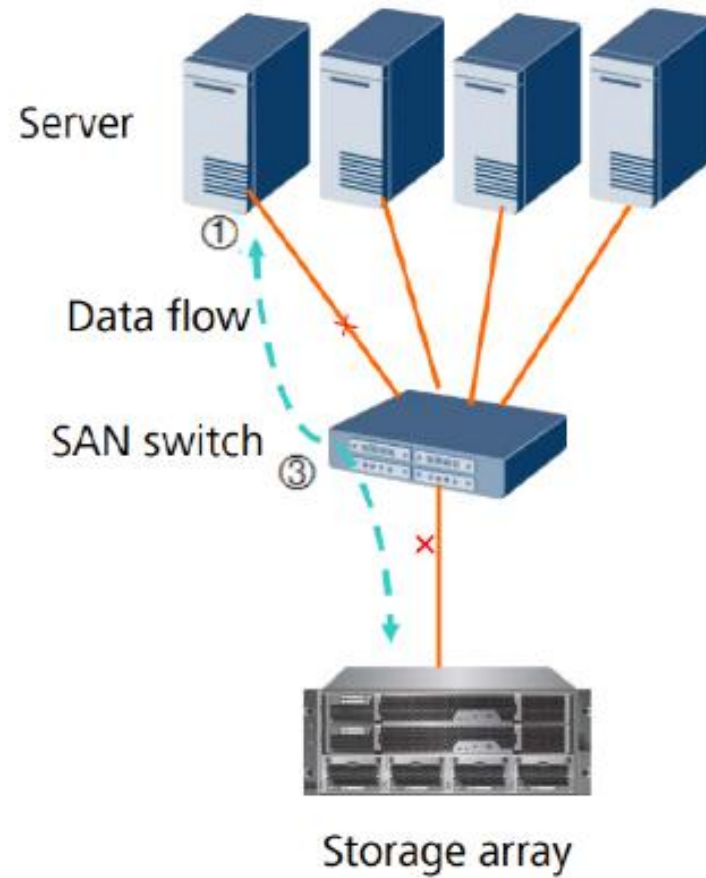
1. Block level storage virtualization

- Block level storage virtualization is implemented in “SAN (storage area network)” and it provides a translation layer in the SAN, between the hosts and the storage arrays. In this type of storage virtualization, servers are redirected to virtualized LUNs instead of LUN's on the individual storage array. These virtualized LUNs are on the virtualized device.

2. File level storage virtualization

- File level storage virtualization happens at NAS (network attached storage) level and help in solving NAS problems by removing the dependencies between the data accessed at the file level and the location where the files are physically stored.

Storage Virtualization





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THANK YOU